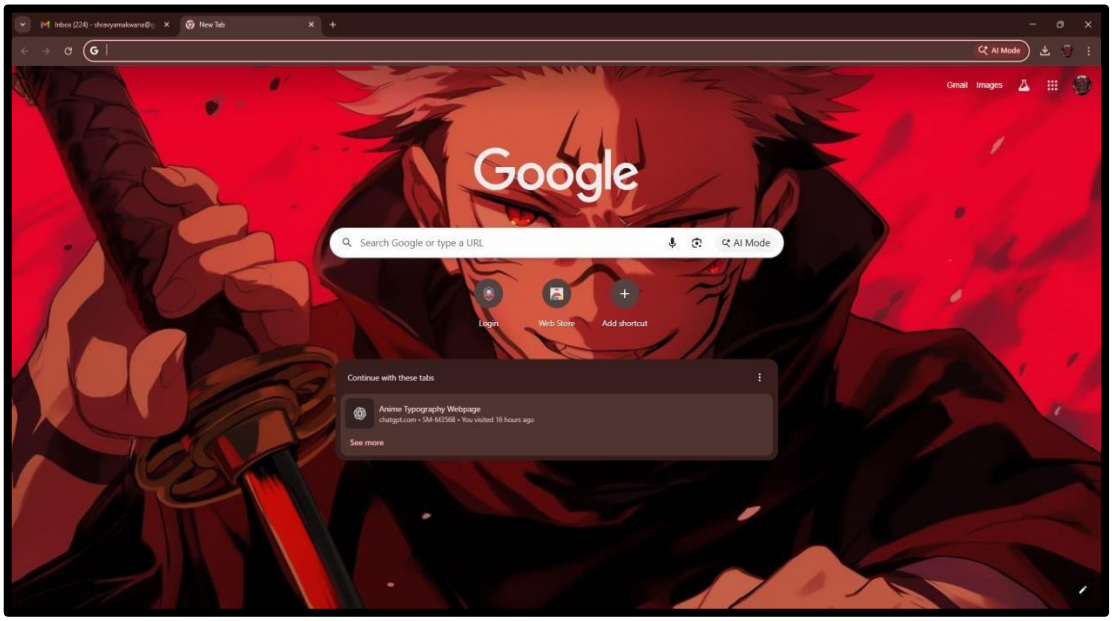
PRACTICAL:1

AIM: Install and configure the Python environment. Run basic Python, commands to verify the Python environment.

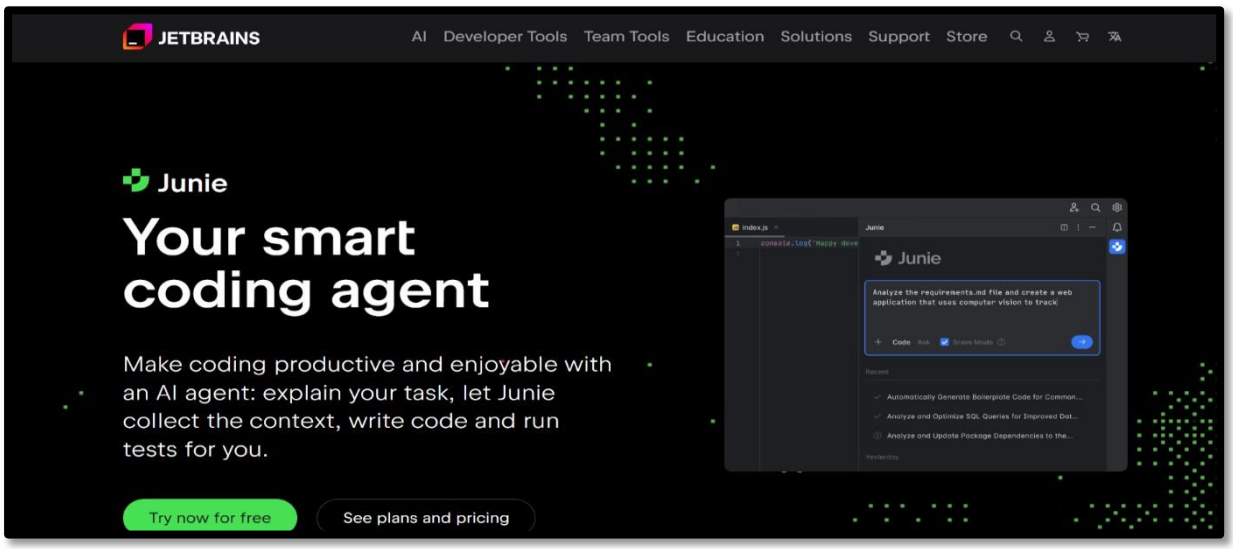
SOLUTION:

Step 1: Download PyCharm

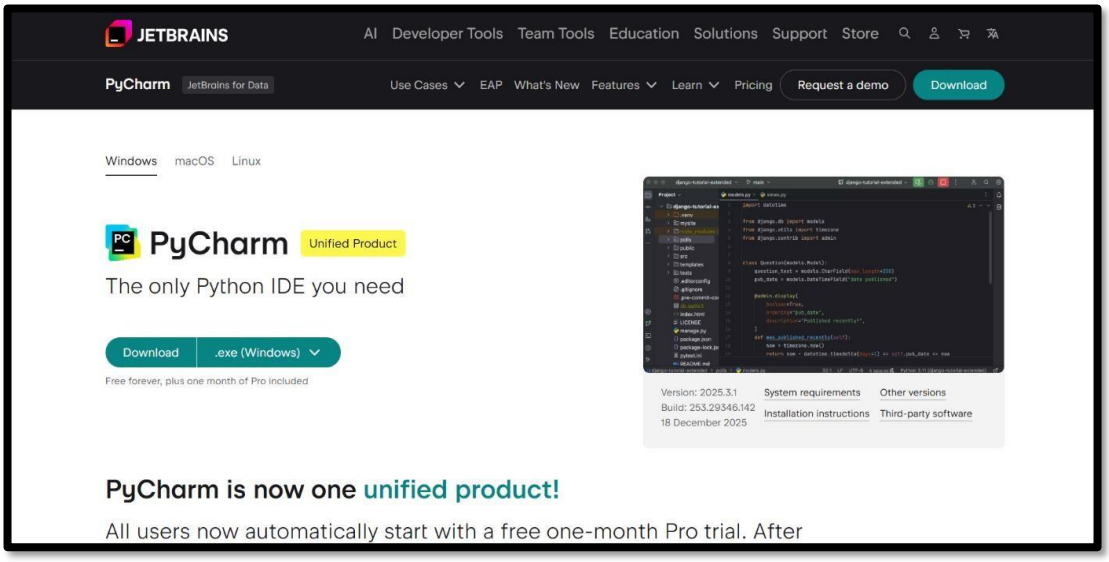
- Open web browser.



- Go to **jetbrains.com/ Pycharm**

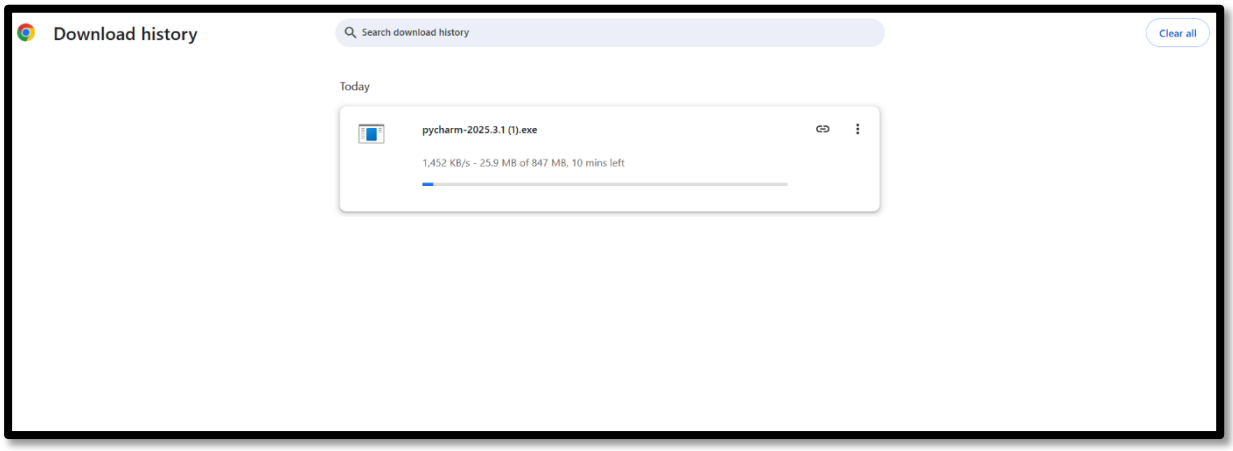


- Click Downlode.



- Select **Community Edition**

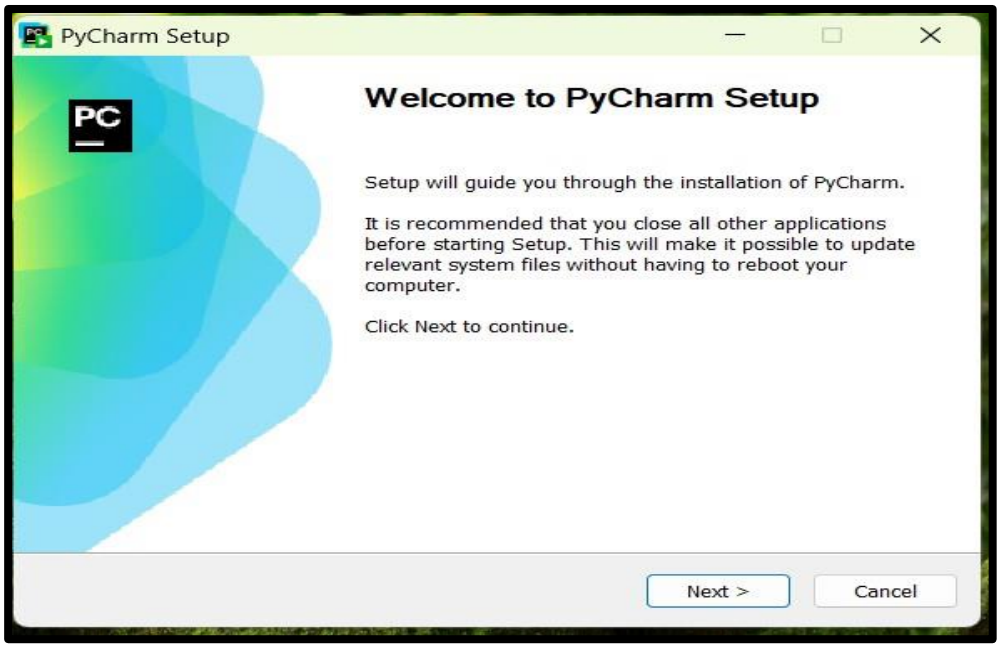
- Download installer for your OS.



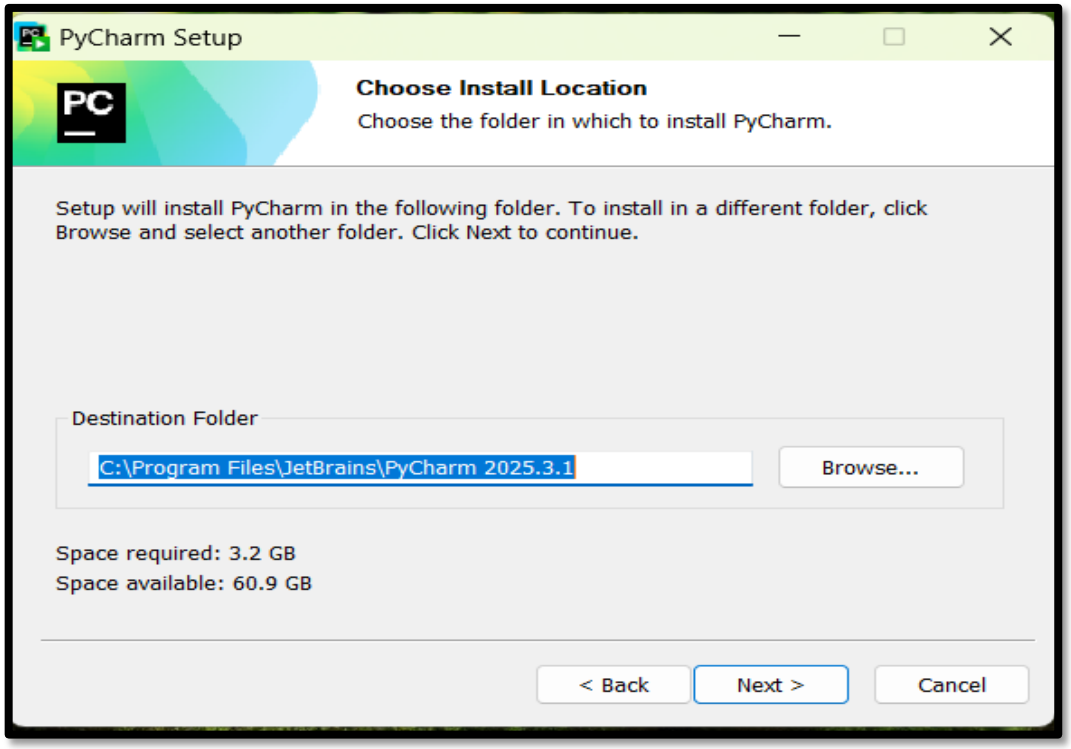
Step 2: Install PyCharm (Windows)

- Run the downloaded installer.

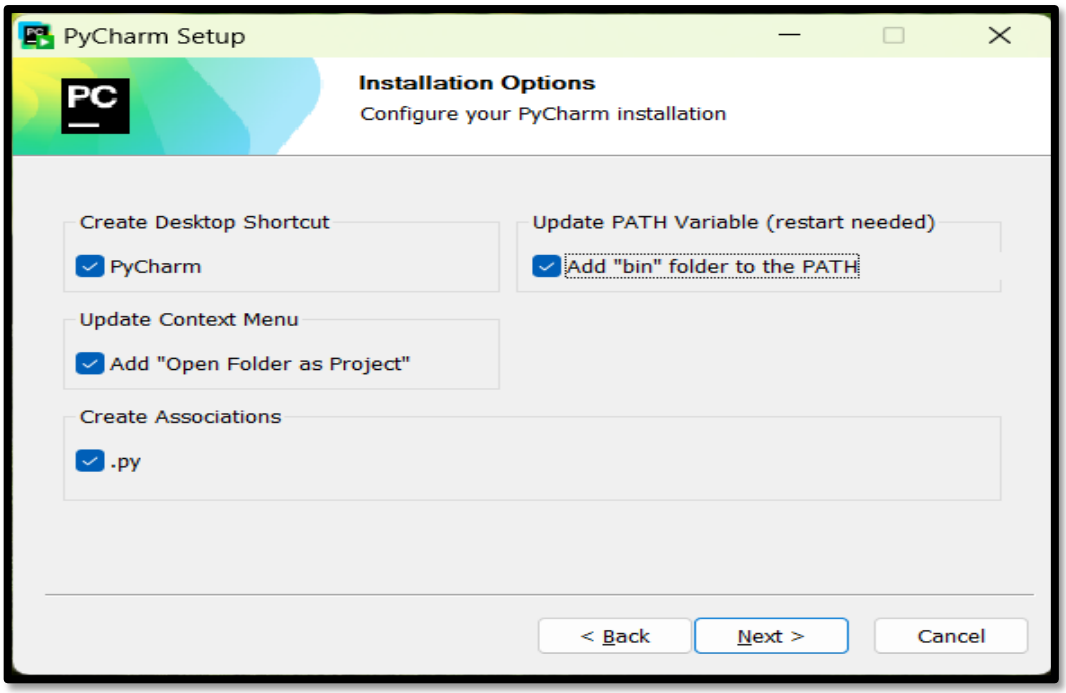
- Click **Next**.



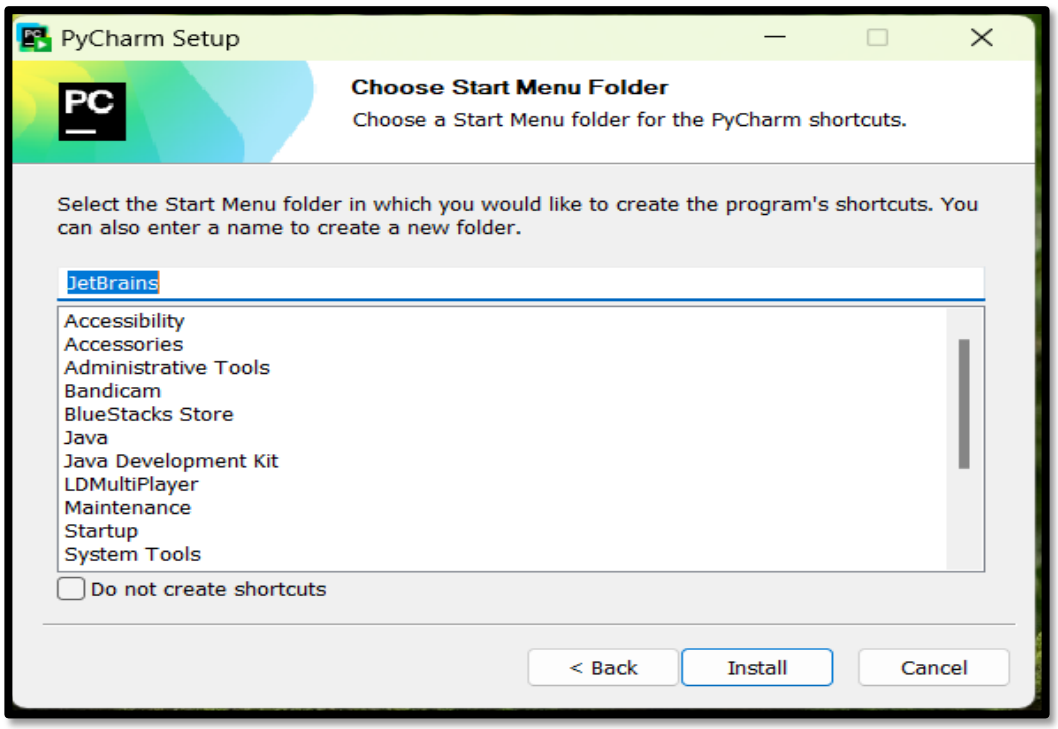
- Select installation path → **Next**



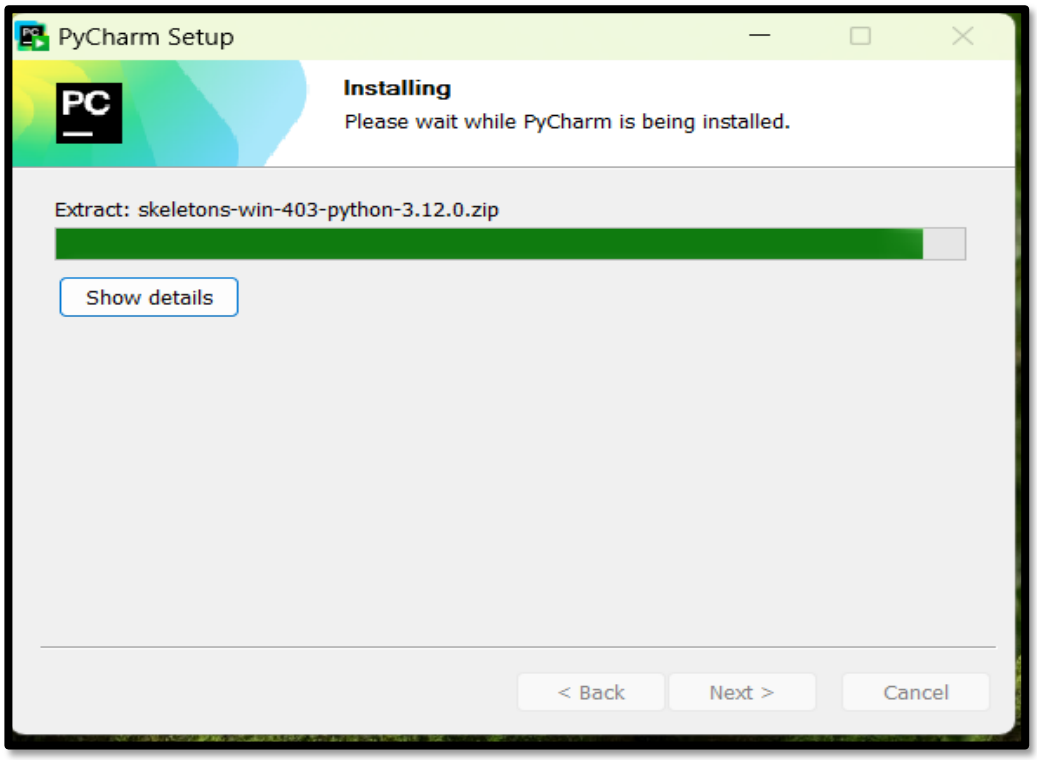
- Check options: Create Desktop Shortcut and Add "Open Folder as Project"



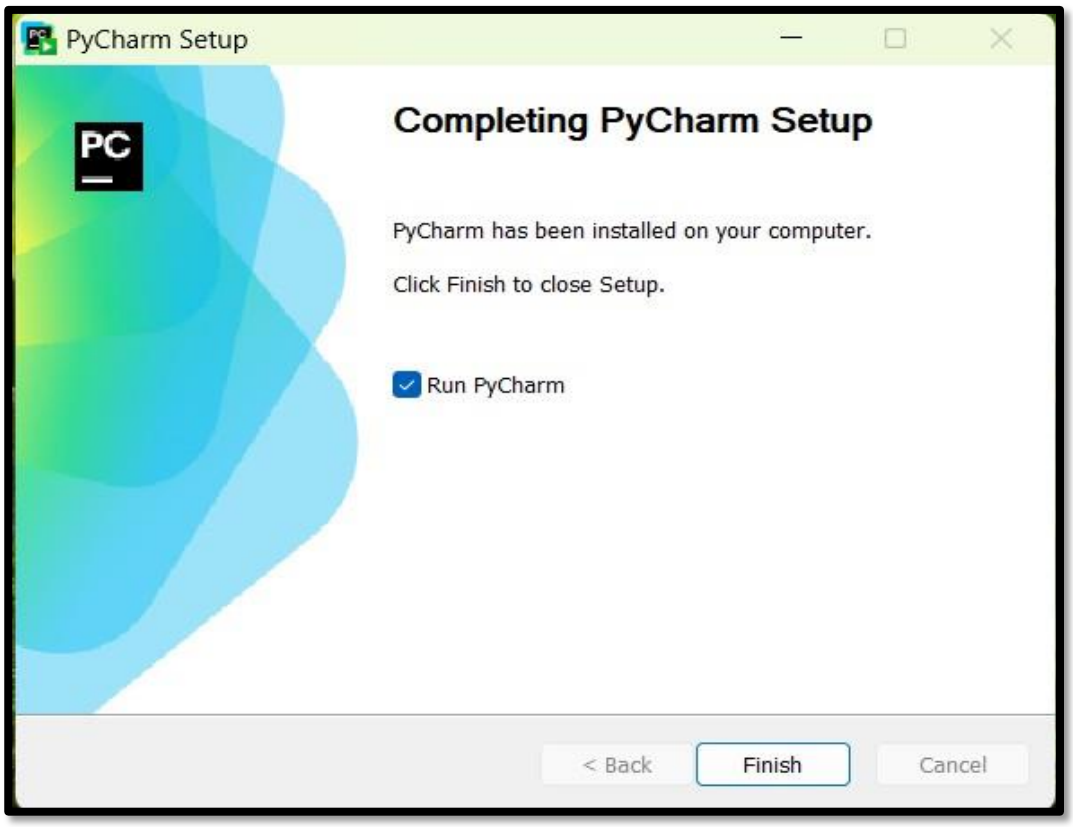
- Add “open folder as project”



- Click **Install**.

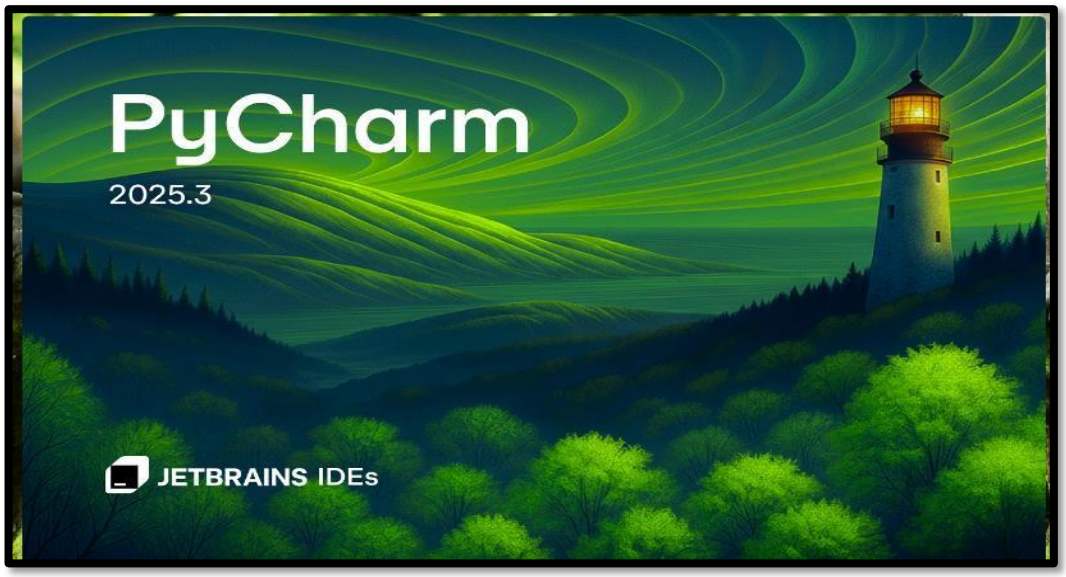


- Finish installation and restart if required.

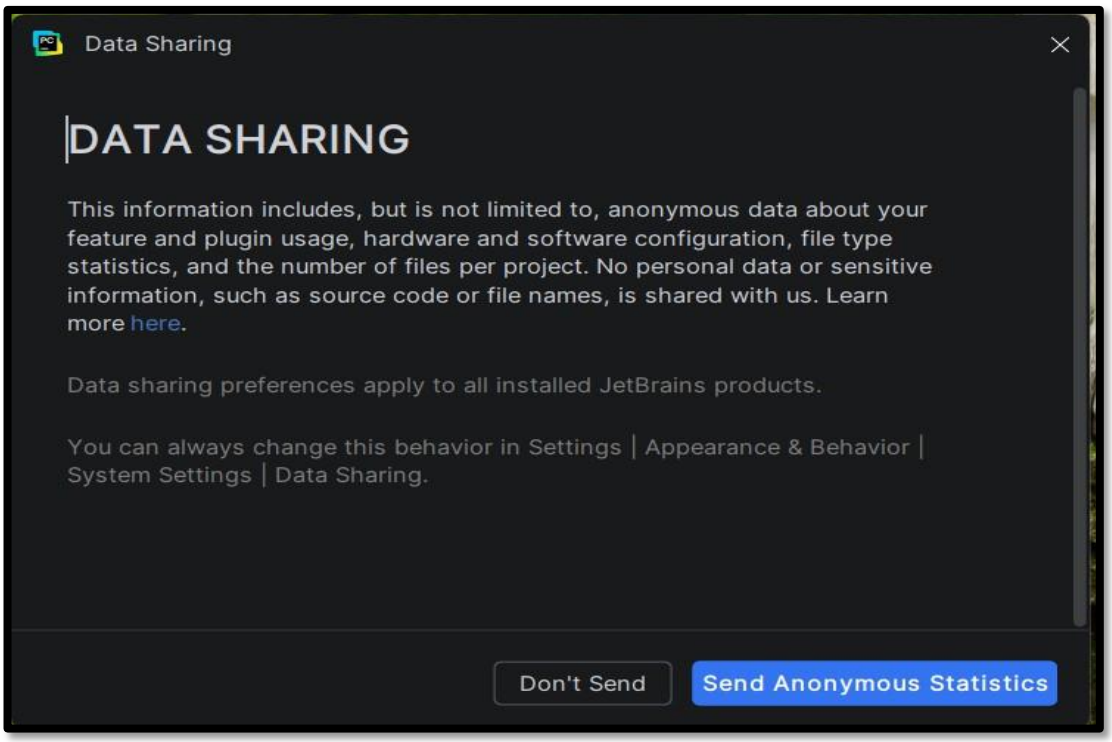


Step 3: Launch PyCharm

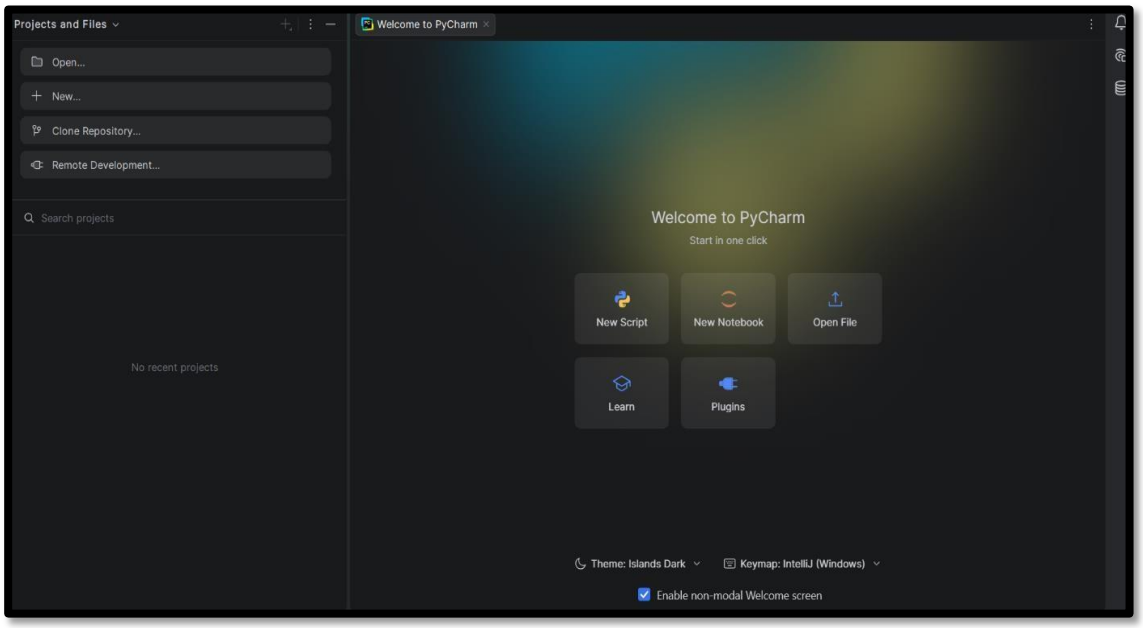
1. Open PyCharm Community Edition.



- Accept license and default settings.



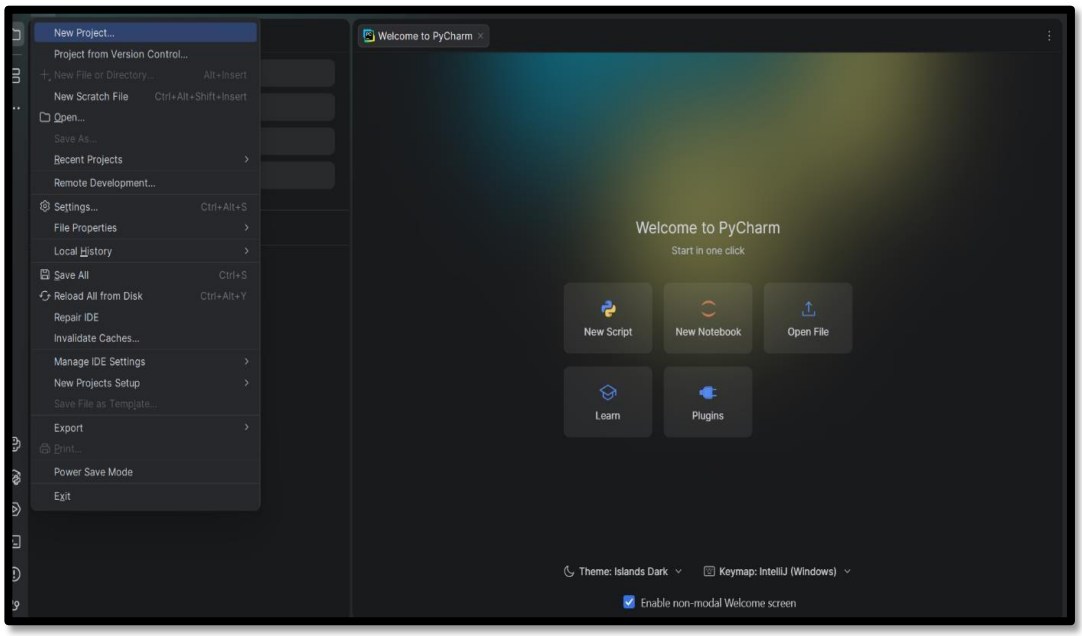
- PyCharm welcome screen will appear.



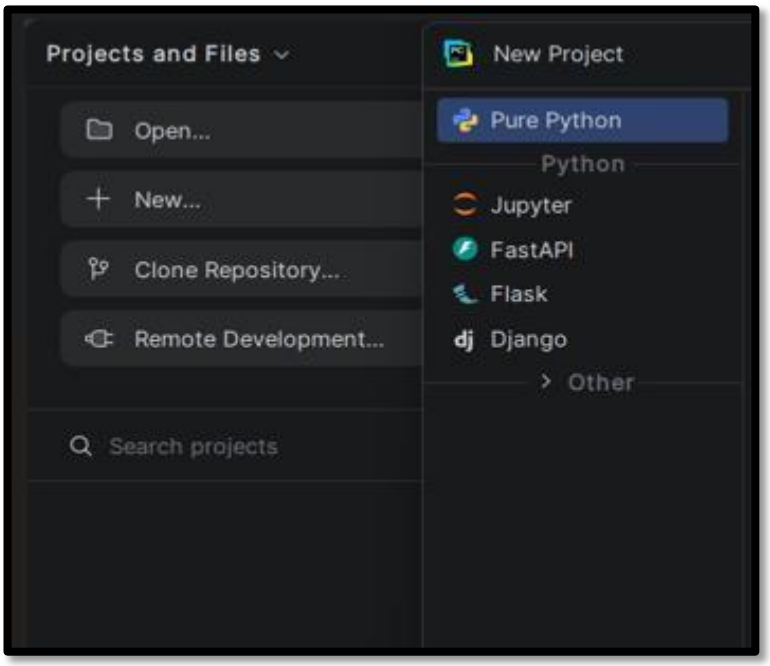
Creating first pycharm program in pycharm

Step:4 create new project

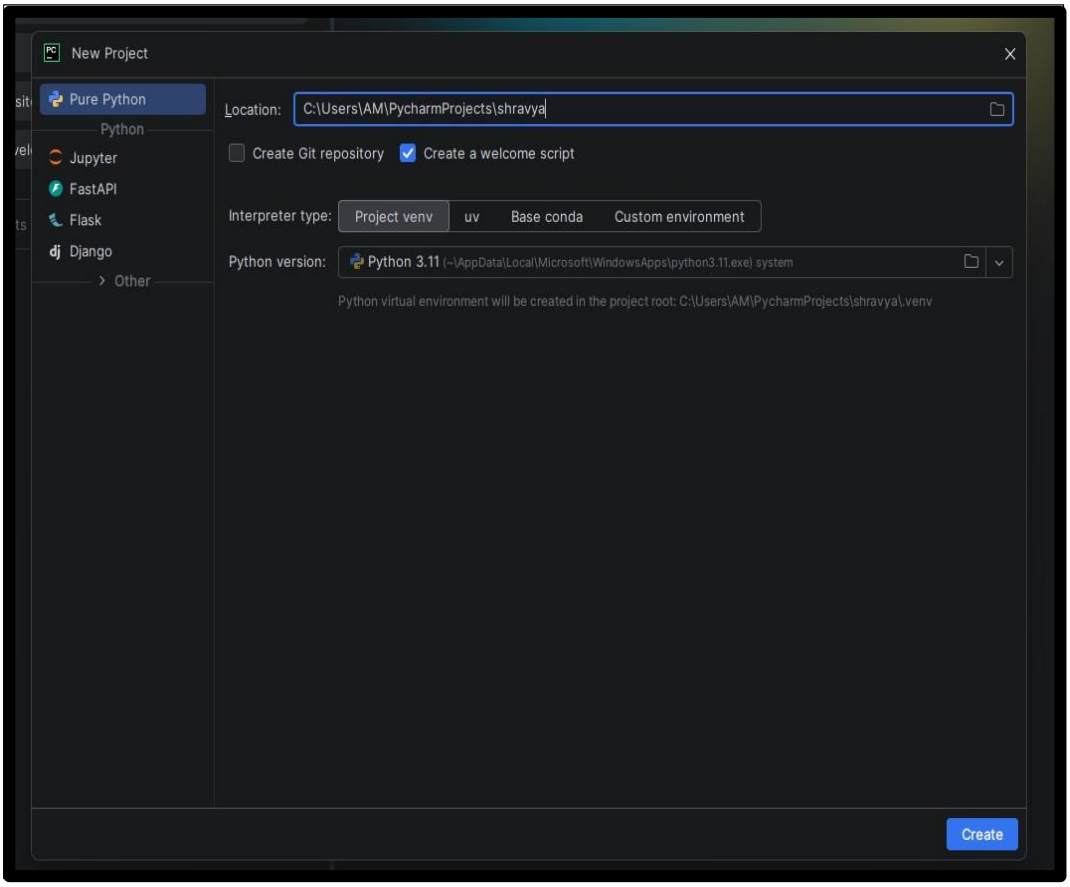
- Click new project.



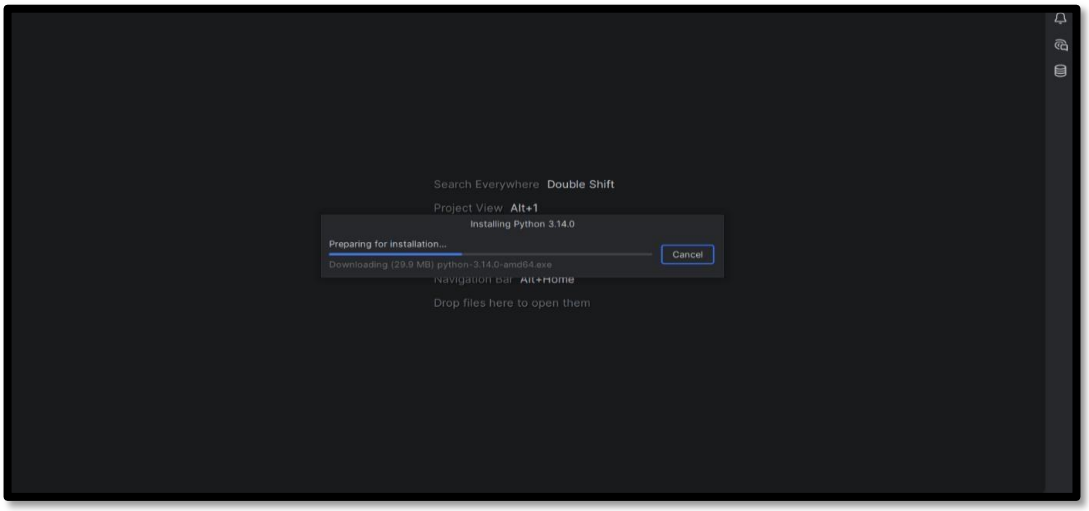
- Select python.



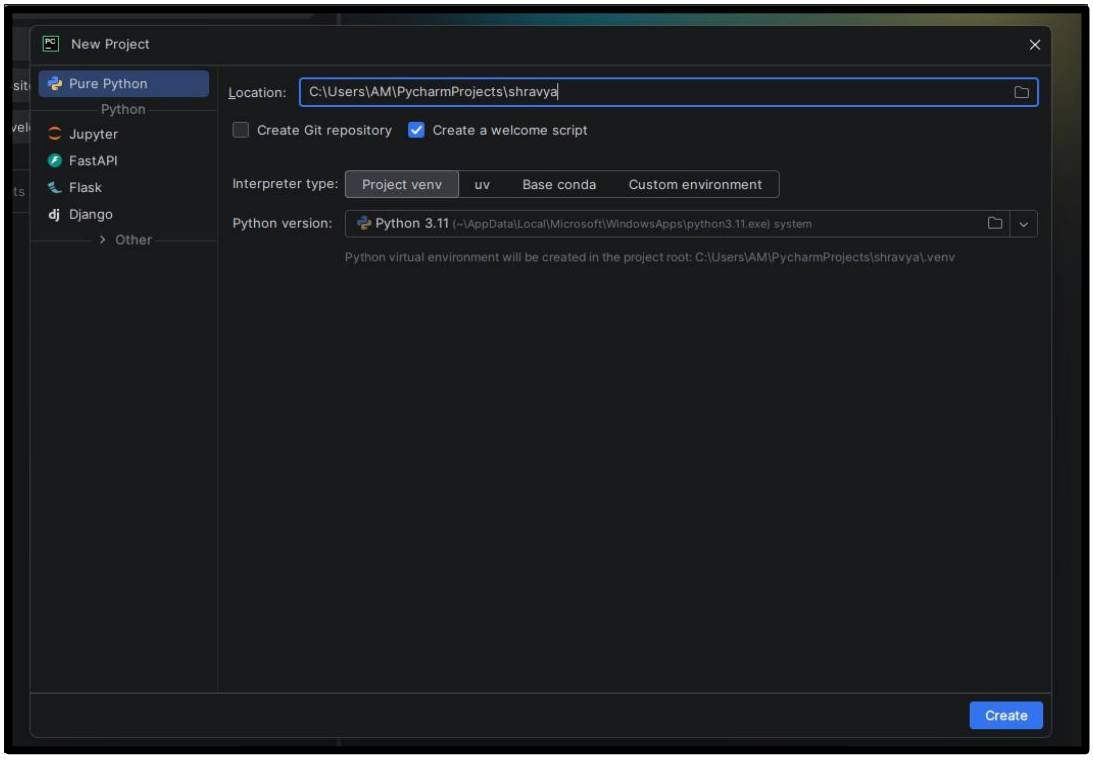
- Choose project location.



- Select existing python interpreter.

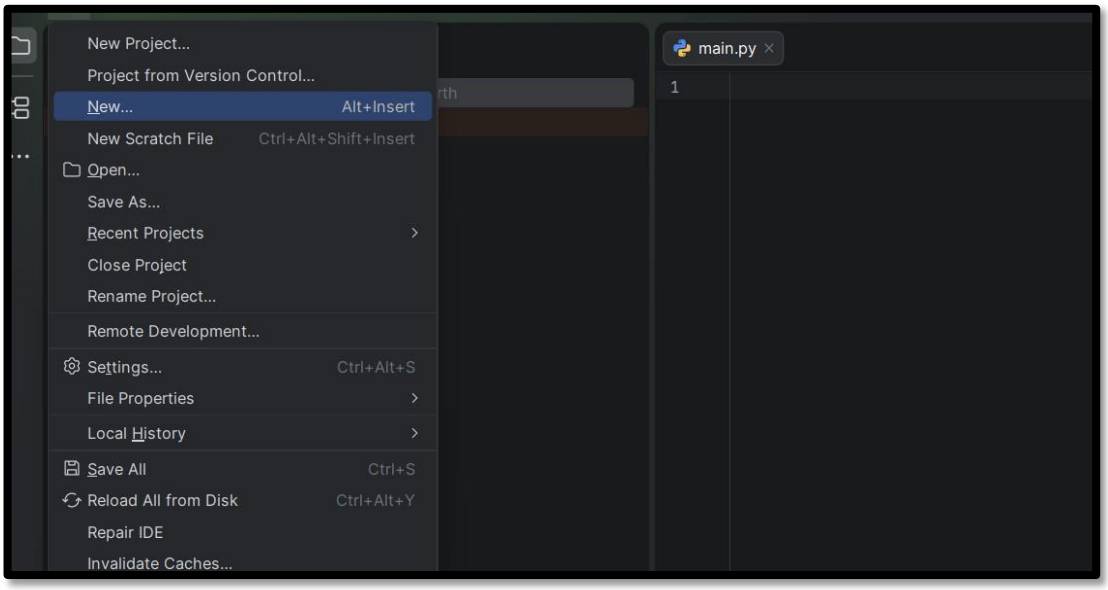


- Click create.

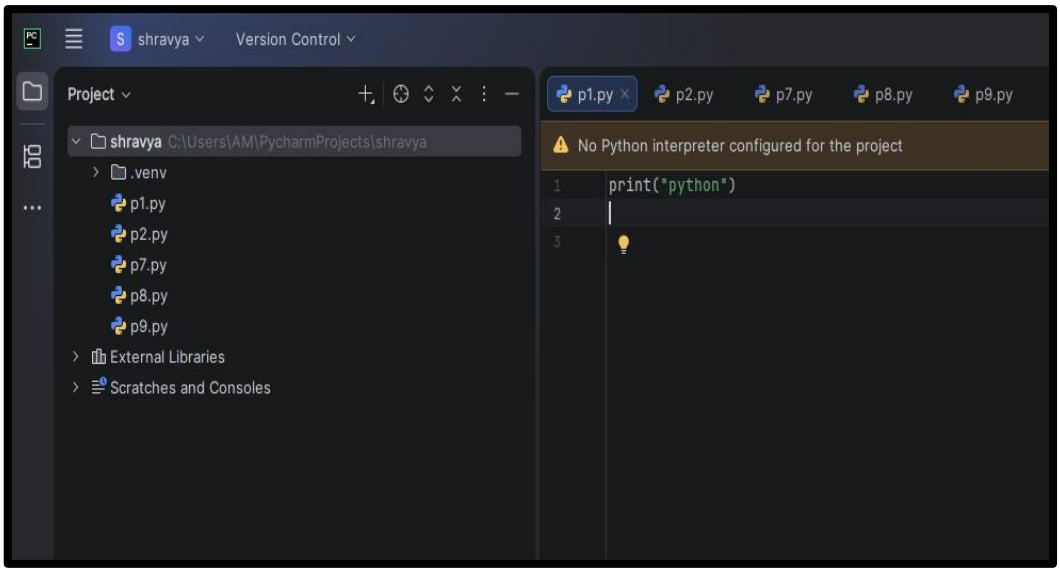


Step:5 create python file.

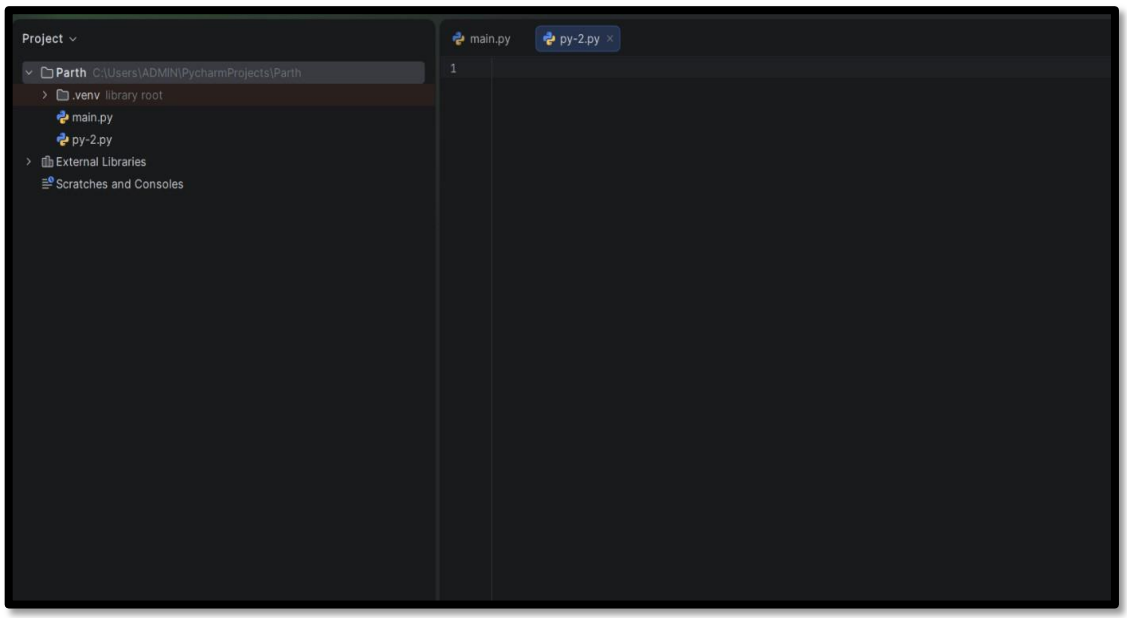
- Right click on project folder.



- Select **New → Python File**

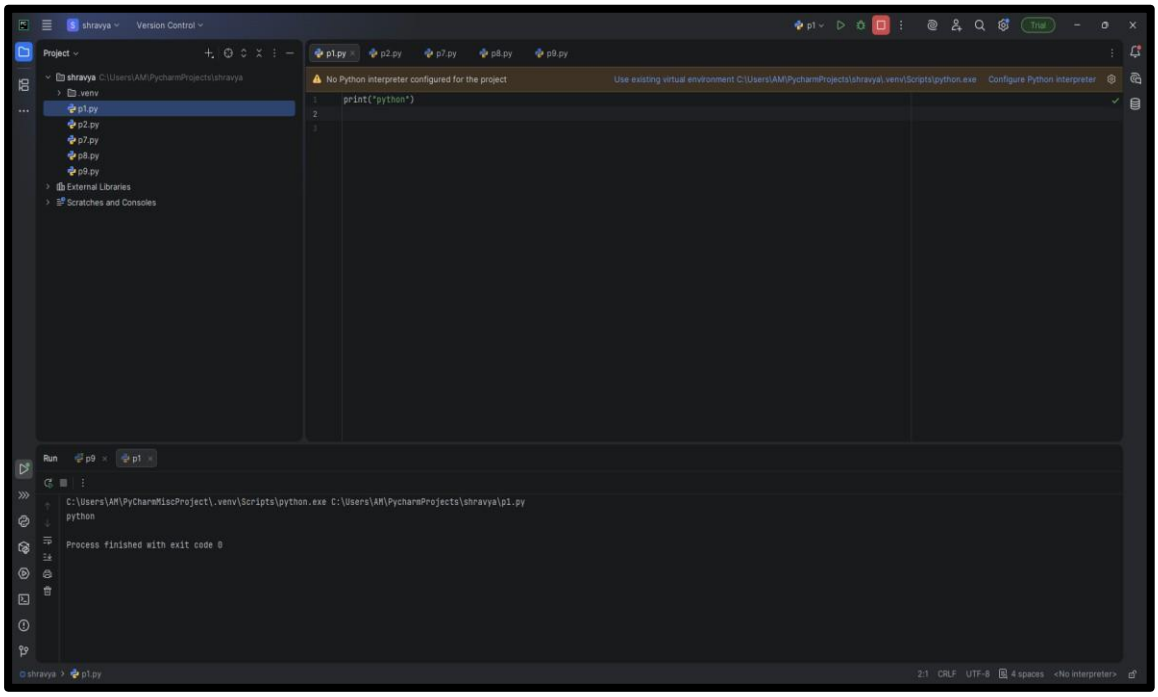


- Enter file name : Practical-1.py



Step 6: Write and Run Program

- Print("python")



Output:

PRACTICAL: 2

AIM: Write a program to read your name, contact number, email, and birthdate and print those details on the screen.

SOLUTION:

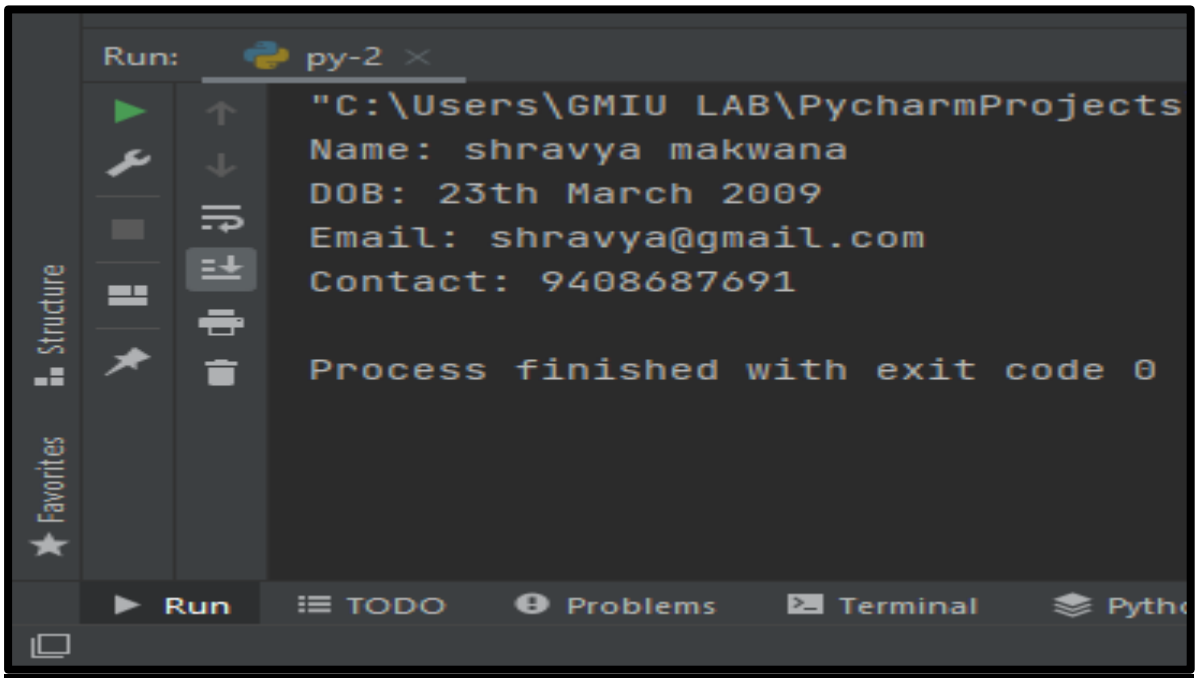
Print("Name: shravya makwana")

Print("DOB: 23th March 2009")

Print("Email: shravya@gmail.com")

Print("Contact: 9408687691")

OUTPUT:



PRACTICAL:3

AIM: Write a program to convert temperature from Celsius to Fahrenheit. Equation to convert Celsius to Fahrenheit: F = (9/5) * C +32.

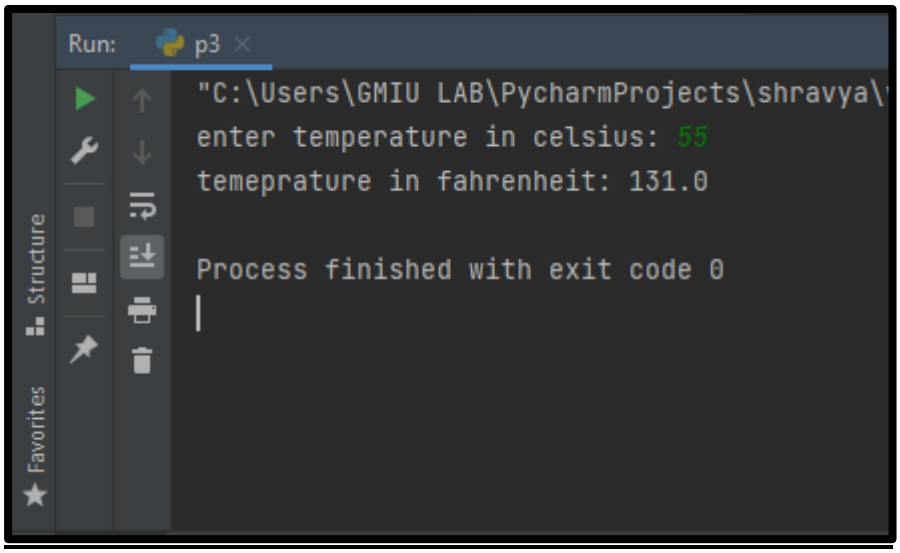
SOLUTION:

```
Celsius = float(input("enter temperature in Celsius: "))

Fahrenheit = (9/5) * Celsius  +  32

Print("temperature in Fahrenheit: ", fahrenheit)
```

OUTPUT:



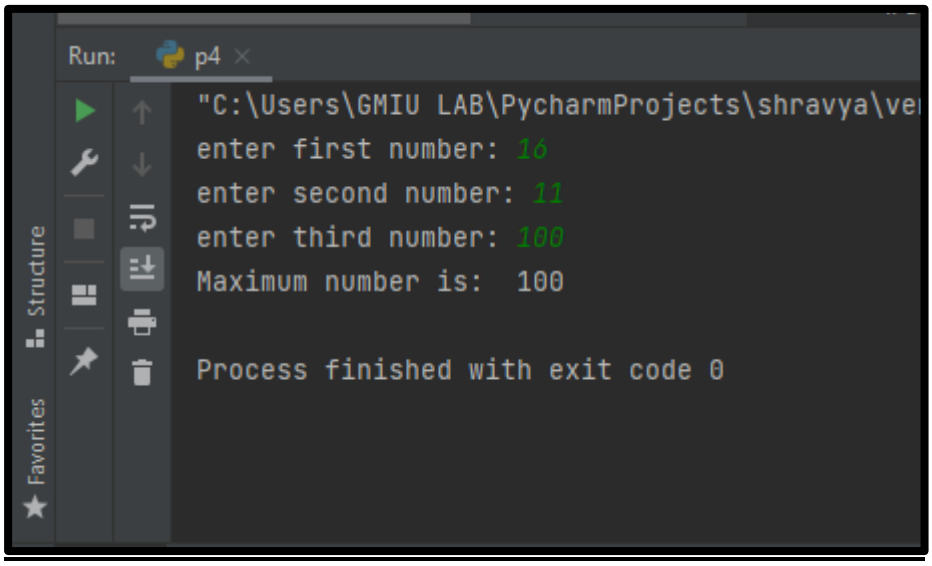
PRACTICAL:4

AIM: Write a program to find a maximum of given three numbers.

SOLUTION:

```
a = int(input("enter first number: "))
b = int(input("enter second number: "))
c = int(input("enter third number: "))
if a > b and a > c:
print("Maximum number is: " , a)
if b > a and a > c:
print("Maximum number is: " , b)
else:
print("Maximum number is: " , a)
```

OUTPUT:



PRACTICAL:5

AIM: A year is a Leap year if it is divisible by 4, unless it is a century year that is not divisible by 400 (1800 and 1900 are not leap years, 1600 and 2000 are leap years). Write a program that calculates whether a given year is a leap year or not.

SOLUTION:

```
year = int(input("Enter a year: ")
if year % 400 == 0: print(year, "is a leap Year")
elif year % 100 == 0: print(year, "is a leap Year")
elif year % 4 == 0: print(year, "is a leap Year")
else: print(year, "is not a leap Year")
```

OUTPUT:

PRACTICAL:6

AIM: Write a program to read the marks and assign a grade to a student, grading system: A (>=90), B (80-89), C (70-79), D (60-69), E (50-59), F (<50), use switch case.

SOLUTION:

```
marks = int(input("Enter a number: ")

grade = marks // 10

match grade:

case 10 | 9:

print("Grade: A")

case 8:

print("Grade: B")

case 7:

print("Grade: C")

case 6:

print("Grade: D")

case 5:

print("Grade: E")

case _:

print("Grade: F")
```

OUTPUT:

```
C:\Users\shahh\PyCharmMiscProject\.venv\Scripts\python.exe "C:\Users\shahh\PyCharmMiscProject\prac 6.py"
Enter student marks: 50
Garde: E

Process finished with exit code 0
```

PRACTICAL:7

AIM: Write a program to read n numbers from users and calculate the average of those n numbers.

SOLUTION:

```
n = int(input("Enter how many numbers you want to enter: "))
total = 0
for i in range(n):
num = float(input(f"enter number {i+1}: "))

 total += num
average = total / n
print("Average =", average)
```

OUTPUT:

PRACTICAL:8

AIM: Write a program that prompts the user to enter 10 integers and displays all the combinations of picking two numbers from the 10.

SOLUTION:

```
numbers = []
print("Enter 5 integers")
for i in range(5):
    num = int(input(f"Enter number {i+1}: "))
    numbers.append(num)
print("\nCombinations of picking two numbers: \n")

for i in range(5):
    for j in range(i + 1, 5):
        print(numbers[i], numbers[j])
```

OUTPUT:

PRACTICAL:9

AIM: Write a program to perform the below operations on the list:

- Create a list.
- Add/Remove an item to/from a list.
- Get the number of elements in the list.
- Access elements of the list using the index.
- Sort the list.
- Reverse the list.

SOLUTION:

```
my_list = []
n = int(input("Enter number of elements: "))
for i in range(n):
    value = int(input("Enter Element: "))
    my_list.append(value)
print("\nList:", my_list)
add = int(input("\nEnter element to add: "))
my_list.append(add)
print("After Adding:", my_list)
remove = int(input("\nEnter element to remove: "))
my_list.remove(remove)
print("After Removing:", my_list)
print("number of elements: ", len(my_list))
index = int(input("\nEnter index to access: "))
print("Element at index", index, "is:", my_list[index])
my_list.sort()
print("Sorted List: ", my_list) my_list.reverse() print("Reversed
List: ",my_list)
```

OUTPUT:

```
Run    p7 ×    p9 ×
C:\Users\AM\PyCharmMiscProject\.venv\Scripts\python.exe C:\Users\AM\PycharmProjects\shravya\p9.py
Enter number of elements: 7
Enter Element: 56
Enter Element: 22
Enter Element: 78
Enter Element: 37
Enter Element: 96
Enter Element: 89
Enter Element: 12

List: [56, 22, 78, 37, 96, 89, 12]
shravya > p9.py
```

```
Run    p7 ×    p9 ×
Enter element to add: 122
After Adding: [56, 22, 78, 37, 96, 89, 12, 122]

Enter element to remove: 37
After Removing: [56, 22, 78, 96, 89, 12, 122]
number of elements:  7

Enter index to access: 2
Element at index 2 is: 78
Sorted List:  [12, 22, 56, 78, 89, 96, 122]
Reversed List:  [122, 96, 89, 78, 56, 22, 12]
shravya > p9.py
```

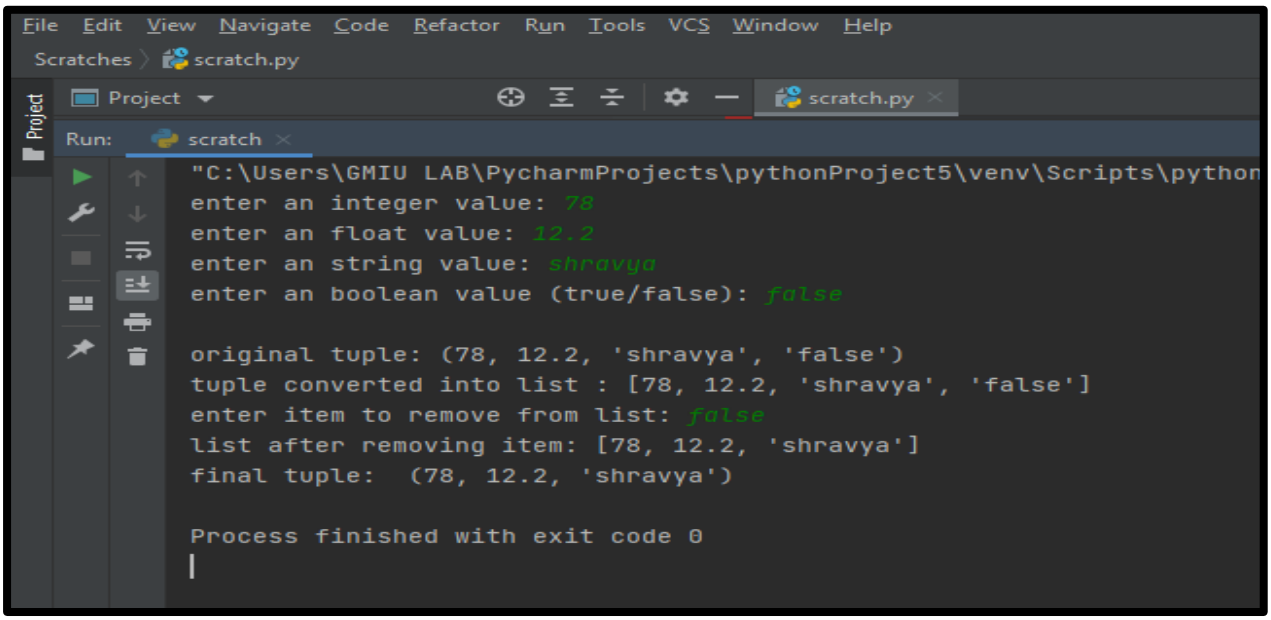
PRACTICAL:10

AIM: Write a program to perform below operations on tuple: Create a tuple with different data types. Print tuple items. Convert tuple into a list. Remove data items from a list. Convert list into a tuple. Print tuple items.

SOLUTION:

```
a = int(input("enter an integer value: "))
b = float(input("enter an integer value: "))
c = str(input("enter an integer value: "))
d = str(input("enter an Boolean  value (true/false): "))
my_tuple = (a, b, c, d)
print("\noriginal tuple:", my_tuple)
my_list = list(my_tuple)
print("tuple converted into list  :", my_list)
remove_item = int("enter item to remove form list: ")
if remove_item.isdigit():
   remove_item = int(remove_item)
elif remove_item.replace('.','.', 1).isdigit() :
     remove_item =  float(remove_item)
my_list remove(remove_item)
print("list after removing item:", my_list)
my_tuple = tuple(my_list)
print("final tuple:  ",my_tuple)
```

OUTPUT:


File Edit View Navigate Code Refactor Run Tools VCS Window Help
Scratches scratch.py
Project
scratch.py
Run: scratch
"C:\Users\GMIU LAB\PycharmProjects\pythonProject5\venv\Scripts\python
enter an integer value: 78
enter an float value: 12.2
enter an string value: shravya
enter an boolean value (true/false): false
original tuple: (78, 12.2, 'shravya', 'false')
tuple converted into list : [78, 12.2, 'shravya', 'false']
enter item to remove from list: false
list after removing item: [78, 12.2, 'shravya']
final tuple: (78, 12.2, 'shravya')
Process finished with exit code 0

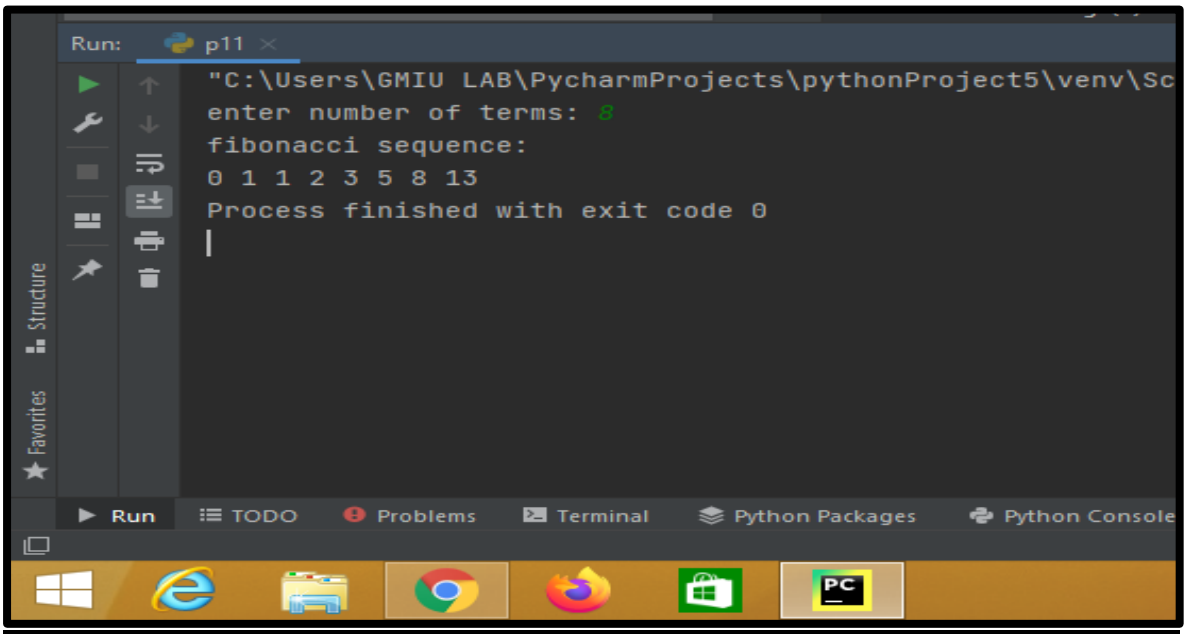
PRACTICAL:11

AIM: Write a program to print Fibonacci sequence up to n numbers using recursion.

SOLUTION:

```
Def  fibonacci(n)  :
      If n == 0:
            return 0
      elif n == 1:
            return 1
      else:
            return Fibonacci(n-1) + Fibonacci(n-2)
n = int(input("enter number of terms:  "))
print("Fibonacci sequence:")
for I in range(n)  :
print Fibonacci(i), end= "")
```

OUTPUT:



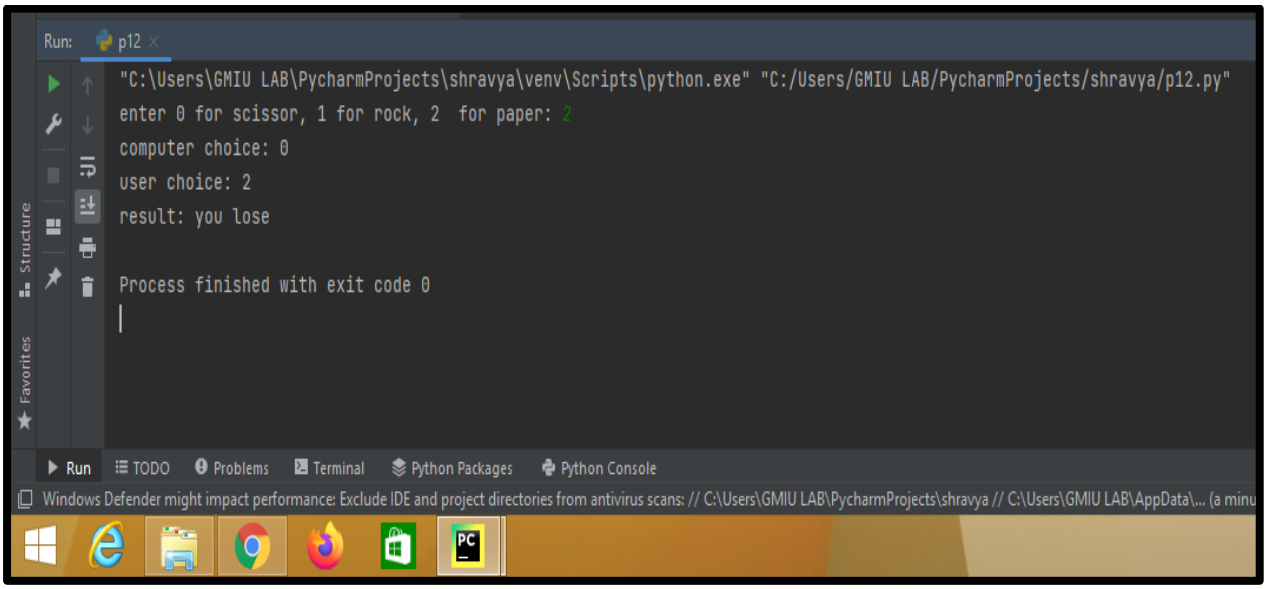
PRACTICAL:12

AIM: Write a program that plays the popular scissor-rock-paper game. (A scissor can cut a paper, a rock can knock a scissor, and a paper can wrap a rock.) The program randomly generates a number 0, 1, or 2 representing scissor, rock, and paper. The program prompts the user to enter a number 0,1, or 2 and displays a message indicating whether the user or the computer wins, loses, or draws.

SOLUTION:

```
import random
computer = random.randint(0, 2)
user = int(input("enter 0 for scissor, 1 for rock, 2  for paper: "))
print("computer choice:", computer)
print("user choice:", user)
if computer == user:
   print("result: draw")
elif (user == 0 and computer == 2) or \
    (user == 1 and computer == 0) or \
    (user == 2 and computer == 1) :
   print("result: you win")
else:
   print("result: you lose")
```

OUTPUT:



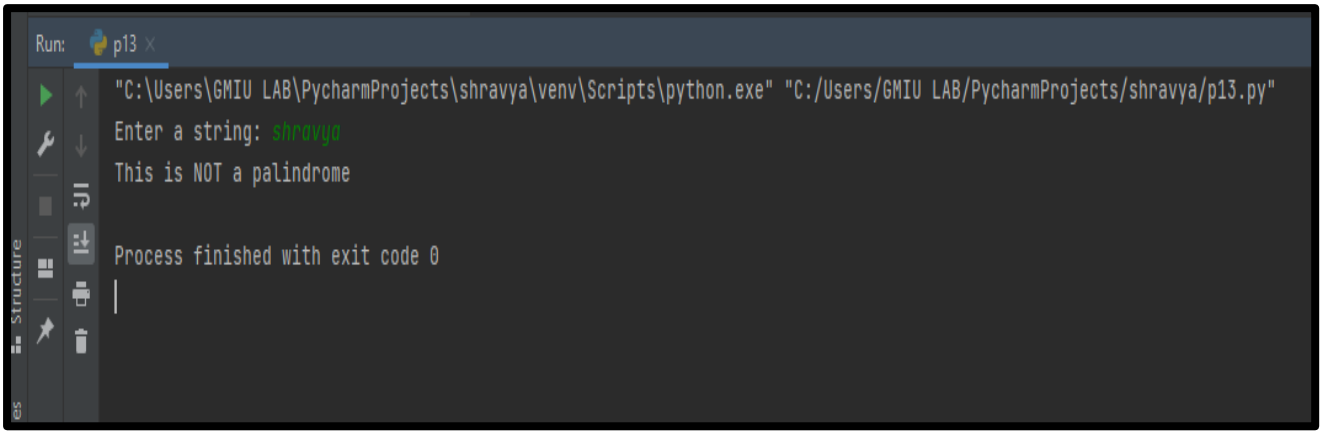
PRACTICAL:13

AIM: Write a program to check whether a given string is palindrome or not.

SOLUTION:

```
string = input("Enter a string: ")
reverse_string = string[::-1]
if string == reverse_string:
    print("The string is a palindrome")
else:
    print("This is NOT a palindrome")
```

OUTPUT:



PRACTICAL:14

AIM: Write a program to perform the below operations on files: Create a text file and write a string to it. Read an entire text file.Read a text file line by line. Write a string to a file. Write a list of strings to a file. Count the number of lines, words in a file.

SOLUTION:

```
file = open("sample.txt", "w")
file.write("python is easy to learn.\n")
file.write("file handling is importent.\n")
file.close()
file = open("sample.txt", "r")
print("reading entire file:")
print(file.read())
file.close()
file = open("sample.txt", "r")
print("reading file line by line:")
for line in file:
    print(line, end="")
file.close()
file = open("sample.txt", "a")
file.write("this is append line.\n")
file.close()
lines = ["line one\n," "line two\n", "line three\n"]
file = open("sample.txt", "a")
file.writelines(lines)
file.close()
file = open("sample.txt", "r")
line_count = 0
word_count = 0
for line in file:
    line_count += 1
    words = line.split()
word_count += len(words)
file.close()
print("\nnumber of lines:", line_count)
print("number of words:", word_count)
```

OUTPUT:

```
Run    p14

C:\Users\AM\AppData\Local\Microsoft\WindowsApps\python3.11.exe
reading entire file:
python is easy to learn.
file handling is importent.

reading file line by line:
python is easy to learn.
file handling is importent.

number of lines: 6
number of words: 2

Process finished with exit code 0
```

PRACTICAL:15

AIM: Write a program to perform the following operations on matrices: Create two matrices, Add the matrices, Subtract the matrices, Multiply the matrices, Transpose the matrices

SOLUTION:

```
# user input for matrix size
r = int(input("Enter number of rows: "))
c = int(input("Enter number of columns: "))

# create first matrix
A = []
print("Enter elements of matrix A:")
for i in range(r):
    row = []
    for j in range(c):
        row.append(int(input()))
    A.append(row)

# create second matrix
B = []
print("Enter elements of matrix B:")
for i in range(r):
    row = []
    for j in range(c):
        row.append(int(input()))
    B.append(row)

print("\nMatrix A:", A)
print("Matrix B:", B)

# Addition
add = []
for i in range(r):
    row = []
    for j in range(c):
        row.append(A[i][j] + B[i][j])
    add.append(row)

print("Addition:", add)

# Subtraction
sub = []
```

```
for i in range(r):
    row = []
    for j in range(c):
        row.append(A[i][j] - B[i][j])
    sub.append(row)

print("Subtraction:", sub)

# Multiplication
mul = [[0 for j in range(c)] for i in range(r)]

for i in range(r):
    for j in range(c):
        for k in range(c):
            mul[i][j] += A[i][k] * B[k][j]

print("Multiplication:", mul)

# Transpose of Matrix A
transpose_A = []

for i in range(c):
    row = []
    for j in range(r):
        row.append(A[j][i])
    transpose_A.append(row)

print("Transpose of Matrix A:", transpose_A)
```

OUTPUT:

```
Run    p15

C:\Users\AM\AppData\Local\Microsoft\WindowsApps\python3.11.exe
Enter number of rows: 2
Enter number of columns: 2
Enter elements of matrix A:
12
45
78
26
Enter elements of matrix B:
14
78
18
45

Matrix A: [[12, 45], [78, 26]]
Matrix B: [[14, 78], [18, 45]]
Addition: [[26, 123], [96, 71]]
Subtraction: [[-2, -33], [60, -19]]
Multiplication: [[978, 2961], [1560, 7254]]
Transpose of Matrix A: [[12, 78], [45, 26]]

Process finished with exit code 0
```

PRACTICAL:16

AIM: Write a program to: Check if a given number is a prime number. 16 Generate all prime numbers within a given range (input from the user).

SOLUTION:

```
num = int(input("Enter a number to check prime: "))

flag = 0

if num > 1:
    for i in range(2, num):
        if num % i == 0:
            flag = 1
            break

if flag == 0 and num > 1:
    print(num, "is a prime number")
else:
    print(num, "is not a prime number")


# Generate prime numbers in a given range

start = int(input("Enter starting range: "))
end = int(input("Enter ending range: "))

print("Prime numbers in the given range:")

for n in range(start, end + 1):
    if n > 1:
        is_prime = True
        for i in range(2, n):
            if n % i == 0:
                is_prime = False
                break
        if is_prime:
            print(n, end=" ")
```

OUTPUT:

```
Run    p16
C:\Users\AM\AppData\Local\Microsoft\WindowsApps\python3.11.exe
Enter a number to check prime: 22
22 is not a prime number
Enter starting range: 2
Enter ending range: 5
Prime numbers in the given range:
2 3 5
Process finished with exit code 0
```

PRACTICAL:17

AIM: Write a program to perform the following operations on a dictionary: Create a dictionary with at least 5 key-value pairs, Add a new key-value pair, Delete a key-value pair, Access elements using a key, Update the value of an existing key, Print all keys, values, and key-value pairs.

SOLUTION:

```
my_dict = {}

print("Enter 5 key-value pairs:")
for i in range(5):
    key = input("Enter key: ")
    value = input("Enter value: ")
    my_dict[key] = value

print("Initial dictionary:", my_dict)

# Add a new key-value pair
key = input("Enter new key to add: ")
value = input("Enter value: ")
my_dict[key] = value
print("After adding:", my_dict)

# Delete a key-value pair
key = input("Enter key to delete: ")
if key in my_dict:
    del my_dict[key]
    print("After deleting:", my_dict)
else:
    print("Key not found!")

# Access element using key
key = input("Enter key to access value: ")
if key in my_dict:
    print("Value:", my_dict[key])
else:
    print("Key not found!")
```

OUTPUT:

```
Run    p17

C:\Users\AM\AppData\Local\Microsoft\WindowsApps\python3.11.exe C:\Users\AM\PycharmProje
Enter 5 key-value pairs:
Enter key: 4
Enter value: 12
Enter key: 45
Enter value: 78
Enter key: 22
Enter value: 14
Enter key: 78
Enter value: 23
Enter key: 14
Enter value: 78
Initial dictionary: {'4': '12', '45': '78', '22': '14', '78': '23', '14': '78'}
Enter new key to add: 100
Enter value: 12
After adding: {'4': '12', '45': '78', '22': '14', '78': '23', '14': '78', '100': '12'}
Enter key to delete: 12
Key not found!
Enter key to access value: 85
Key not found!

Process finished with exit code 0
```